

Sea Turtle Re-Identification with DINOv3 Vision Transformers

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Abstract—Sea turtle populations continue to decline, and conservation programmes need to identify individuals without the cost and invasiveness of satellite tagging. We present a re-identification (Re-ID) system for sea turtle head images built on a Vision Transformer (ViT) backbone pretrained with DINOv3, trained under a strict open-set identity split on the SeaTurtleIDHeads dataset. A baseline configuration derived from literature conventions reaches 78.3% Rank-1 and 84.9% mAP; a tuned configuration reaches 90.4% Rank-1 and 93.5% mAP. A one-factor-at-a-time study over 19 parameters shows that the backbone learning rate alone accounts for roughly 78% of the total improvement, and that individual gains are strongly sub-additive: summing them predicts an arithmetically impossible 112% Rank-1, while only 36% of the predicted gain is realised when combined. Applied to SeaTurtleID2022 without any retraining, the model retains 81.1% Rank-1 and 97.2% Rank-5, confirming that the learned embedding transfers across capture domains. The system’s real limitation appears only when unregistered individuals must be rejected: Registered Rank-1 falls to 55.3% with a global threshold and 56.7% with a background class, both at a false-reject rate near 42%. Two fundamentally different decision rules converge on nearly the same operating point because top-1 cosine scores of registered and unregistered queries are confined to a narrow band (ROC-AUC 0.77–0.79). In the worst failures the correct identity still ranks first and is discarded by the rejection stage. The bottleneck is therefore score calibration, not representation quality.

Index Terms—Sea turtle, re-identification, Vision Transformer, DINOv3, open-set recognition, wildlife monitoring

I. INTRODUCTION

Sea turtles are among the most ecologically significant yet threatened groups of marine reptiles. Indonesia alone hosts six of the world’s seven species [1]. Populations have declined sharply due to coastal development that destroys nesting habitats and to illegal egg collection and poaching [1]. Documented cases in West Sulawesi between 2016 and 2019 reported more than twenty turtles killed for their eggs and meat [2].

Effective conservation requires tracking individuals non-invasively over long periods in order to estimate population

size, study migration, and monitor reproduction. The conventional approach, satellite tagging, attaches a GPS transmitter to the animal. It is accurate but expensive, vulnerable to device malfunction, and imposes stress during attachment [3].

Computer-vision re-identification (Re-ID) is a non-invasive alternative. Re-ID recognises a previously observed individual from new images using unique visual characteristics [4]. For sea turtles the scale (scute) patterns on the head remain stable throughout an individual’s lifetime [5], functioning as a natural biometric. Vision Transformers [13] combined with self-supervised pretraining such as DINOv3 [14] produce rich, transferable representations well suited to this fine-grained task.

Most wildlife Re-ID work reports closed-gallery retrieval accuracy. Field deployment, however, is inherently open-set in a second and harder sense: a photograph may show an animal that has never been registered, and the system must say so. We find that this distinction dominates practical performance. Our contributions are:

- A DINOv3 ViT Re-ID pipeline for sea turtle head images, trained under an open-set identity split, reaching 90.4% Rank-1 and 93.5% mAP on held-out identities.
- A sensitivity map over 19 parameters showing that the backbone learning rate is the single dominant factor, and that per-parameter gains measured by one-factor-at-a-time (OFAT) tuning are sub-additive and cannot be summed.
- A cross-dataset evaluation on SeaTurtleID2022 without retraining, quantifying the cost of domain change at 9.3 percentage points (pp) of Rank-1.
- Evidence that under unregistered-identity rejection the limiting factor is score calibration rather than embedding quality: two structurally different rejection rules reach nearly identical operating points, and correct matches remain rank-1 even among the most confident errors.

II. RELATED WORK

A. Sea Turtle Individual Identification

Early automated systems used neural network ensembles on facial scute images of green turtles [6]. Carpentier et al. [5] validated the long-term stability of facial scute patterns over 11 years, confirming their suitability as persistent biometric markers. Alternative body regions such as front-flipper scales have also been explored [7], [8], with some showing higher accuracy under specific conditions.

Adam et al. [9] introduced SeaTurtleID2022, a 13-year longitudinal dataset of 438 loggerhead turtles, together with a time-aware open-set protocol and a baseline integrating Hybrid Task Cascade segmentation with ArcFace feature extraction, reaching 86.8% Rank-1. Nguyen et al. [10] evaluated AlexNet, GoogLeNet, VGG-19, and ResNet-50 on facial scute images, with ResNet-50 reaching 95.3% accuracy on a mixed dataset under a closed-set protocol. Čermák et al. [11] introduced WildlifeDatasets and MegaDescriptor, the first multi-species Re-ID foundation model, outperforming general pretrained models including CLIP and DINOv2. Papafitsoros et al. [12] improved accuracy further by exploiting left-right facial profile similarities within the same individual.

B. Vision Transformers and Metric Learning

ViT [13] applies the Transformer architecture directly to sequences of image patches; self-attention captures global spatial relationships in a single layer, unlike the bounded receptive field of convolutions. DINO [16], DINOv2 [15], and DINOv3 [14] provide self-supervised ViT features that transfer strongly to downstream recognition without labels. Re-ID itself is usually framed as metric learning [19], with ArcFace [17] enforcing an additive angular margin between class prototypes and batch-hard triplet mining [18] shaping local distances. K-reciprocal re-ranking [20] is a standard unsupervised post-processing step.

III. METHODOLOGY

A. Dataset and Protocol

SeaTurtleIDHeads [25] contains 7,582 images of 400 individual sea turtles collected over more than a decade. All images are cropped to the head region and vary widely in pose, lighting, camera angle, distance, and resolution, reflecting real field conditions. Head detection and segmentation are outside the scope of this work; inputs are assumed pre-cropped.

The dataset is split into training (70%), validation (20%), and test (10%) partitions by an open-set identity split: no individual appears in more than one partition, so every reported result concerns identities never seen during training. Identities with fewer than $N_G + 2$ images are excluded, guaranteeing N_G gallery images and at least two query images per identity. Within each evaluation partition, $N_G = 3$ images per identity are randomly assigned to the gallery and mean-pooled into a single ℓ_2 -normalised prototype; the remainder form the query set. Validation is used only for checkpoint selection; the test partition is used once.

B. Preprocessing and Augmentation

Images are resized to 256×256 . Training augmentations are applied sequentially: random crop (resize to 288×288 , crop to 256×256), random horizontal flip ($p = 0.5$), colour jitter over brightness, contrast, saturation and hue, random grayscale ($p = 0.1$), random rotation up to 20° , Gaussian blur (kernel 5×5 , $\sigma \in [0.1, 2.0]$), and random erasing ($p = 0.3$) to simulate partial occlusion by seagrass or shadow. At evaluation only resize and normalisation are applied. Pixels are normalised with ImageNet statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$).

C. Model Architecture

1) *Backbone*: The baseline backbone is ViT-B/16 pretrained with DINOv3 [14], loaded via the PyTorch Image Models (timm) library. A 256×256 input yields 256 patch tokens; after 12 Transformer blocks, the 768-dimensional [CLS] token serves as the global image representation.

2) *Embedding Head*: The backbone output passes through Layer Normalisation, dropout (p_{drop}), and a bias-free linear projection $768 \rightarrow d_e = 256$. The output is ℓ_2 -normalised, placing all embeddings on the unit hypersphere so that cosine similarity reduces to a dot product.

D. Loss Functions

ArcFace [17] adds an angular margin m_a to the angle between an embedding and its correct class prototype before softmax cross-entropy:

$$\mathcal{L}_{\text{ArcFace}} = -\frac{1}{N} \sum_{i=1}^N \log \frac{e^{s_a \cos(\theta_{y_i} + m_a)}}{e^{s_a \cos(\theta_{y_i} + m_a)} + \sum_{j \neq y_i} e^{s_a \cos \theta_j}} \quad (1)$$

with scale $s_a = 32.0$ and margin $m_a = 0.5$. The scale is set below the value of 64 used in the original formulation because the identity count here is far smaller than in large-scale face benchmarks; Section IV-C confirms this choice empirically.

Online Hard Triplet Loss [18] mines the hardest positive and hardest negative within each batch:

$$\mathcal{L}_{\text{Triplet}} = \frac{1}{N} \sum_{i=1}^N \max(d(a_i, p_i) - d(a_i, n_i) + m_t, 0) \quad (2)$$

where $d(\cdot, \cdot)$ is the Euclidean distance and $m_t = 0.3$. The total objective is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{ArcFace}} + \lambda_t \cdot \mathcal{L}_{\text{Triplet}}. \quad (3)$$

E. Training Configuration

Each batch contains $P = 16$ identities with $K = 4$ images each (batch size 64), sampled with replacement so that valid positive pairs always exist. AdamW [22] is used with weight decay $\lambda_w = 10^{-4}$. Layer-wise Learning Rate Decay (LLRD) [23] assigns each Transformer block i of $L = 12$ its own rate:

$$\eta_i = \eta_{bb} \cdot \gamma^{L-i}, \quad \gamma = 0.75 \quad (4)$$

with backbone base rate $\eta_{bb} = 10^{-5}$, while the randomly initialised embedding head and ArcFace classifier use $\eta_{base} = 10^{-3}$. A linear warmup of $W = 5$ epochs precedes cosine decay [24] over $E = 50$ epochs:

$$\eta(e) = \begin{cases} \eta_{max} \cdot \frac{e+1}{W} & e < W \\ \frac{\eta_{max}}{2} \left(1 + \cos \left(\pi \cdot \frac{e-W}{E-W} \right) \right) & e \geq W \end{cases} \quad (5)$$

Gradients are clipped at $G_{max} = 1.0$; validation runs every 5 epochs with early stopping patience of 10 intervals; automatic mixed precision is enabled.

F. Inference and Evaluation

Embeddings are extracted with a single forward pass over the resized image and ℓ_2 -normalised; no test-time augmentation is used, so all reported differences are attributable to training and to domain, not to inference protocol. Similarity is cosine, optionally refined by k-reciprocal re-ranking [20]:

$$D_{final}(q, g) = \lambda_{rr} D_{cosine}(q, g) + (1 - \lambda_{rr}) D_{Jaccard}(q, g) \quad (6)$$

with $k_1 = 20$, $k_2 = 6$, $\lambda_{rr} = 0.3$. We report Rank-1/5/10 from the Cumulative Matching Characteristic and mean Average Precision (mAP) [21].

IV. EXPERIMENTS

A. Experimental Setup

All experiments ran on a single NVIDIA GeForce RTX 4090 (24 GB) with an Intel Core i9-13900KF, 64 GB RAM, Windows 11, Python 3.10, PyTorch 2.11.0, and timm 1.0.25. A fixed seed (42) was used throughout; variance across seeds was not measured, which bounds the conclusions drawn from small effects (Section V).

B. Baseline Results

The baseline configuration in Table I was fixed from literature conventions *before* any variation was run, so that the OFAT analysis is not circular: a parameter’s contribution can only be judged fairly if the reference point was chosen without using the results it is meant to evaluate. Table II reports its performance.

Accuracy rises steeply from Rank-1 to Rank-5 (+14.0 pp) and only marginally from Rank-5 to Rank-10 (+2.8 pp). For roughly 14% of queries the correct identity is already within the top five but loses narrowly at rank 1, so most failures are ordering errors rather than retrieval failures. Re-ranking helps at the top of the list and on mAP while slightly hurting Rank-10, which is the expected trade-off for a method that optimises nearest-neighbour consistency.

TABLE I
BASELINE CONFIGURATION

Category	Parameter	Value
Dataset	Image size	256
	Gallery images / identity N_G	3
Architecture	Backbone	ViT-B/16 DINOv3
	Embedding dim d_e	256
	Dropout p_{drop}	0.1
Loss	ArcFace scale s_a	32.0
	ArcFace margin m_a	0.5
	Triplet margin m_t	0.3
	Triplet weight λ_t	0.5
Training	Identities / batch P	16
	Images / identity K	4
	Epochs E	50
	Warmup W	5
Optimiser	Weight decay λ_w	10^{-4}
	Base LR η_{base}	10^{-3}
	Backbone base LR η_{bb}	10^{-5}
	LLRD factor γ	0.75

TABLE II
BASELINE EVALUATION ON SEATURTLEIDHEADS TEST IDENTITIES

Metric	Cosine	+Re-rank	Δ (pp)
Rank-1 (%)	77.4	78.3	+0.9
Rank-5 (%)	91.5	92.3	+0.8
Rank-10 (%)	95.9	95.1	−0.8
mAP (%)	83.6	84.9	+1.3

C. Parameter Sensitivity

Nineteen parameters were varied one at a time with all others held at baseline. Table III summarises the outcome, ordered by Rank-1 range, i.e. the spread between the best and worst tested value. Range measures how much performance moves when a parameter is misconfigured, independently of whether the baseline was already optimal; Δ measures the headroom available above the baseline’s 78.3%.

Three groups emerge. **Sensitive but already optimal:** η_{base} , s_a , and image size cannot be improved upon, yet misconfiguring them is costly. Moving η_{base} by one decade in either direction costs more than 22 pp of Rank-1, because the randomly initialised head either fails to converge or never learns within the epoch budget. **Real headroom:** η_{bb} , E , λ_t , p_{drop} , backbone, m_a , and W . Raising the backbone rate from 10^{-5} to 10^{-4} yields 87.8% Rank-1 and 91.6% mAP on its own, showing that the conventional, conservative fine-tuning rate was the single largest constraint on the system. Since the trend is monotone and 10^{-4} is the upper bound tested, the true optimum may lie higher. **Negligible:** the bottom six parameters move Rank-1 by under 2 pp, which on a query set of a few hundred amounts to a handful of queries and is not separable from seed noise.

Two results contradict common practice and deserve emphasis. First, disabling Triplet Loss entirely ($\lambda_t = 0$) is the

TABLE III
PARAMETER SENSITIVITY, ORDERED BY RANK-1 RANGE (+RE-RANK)

Param	Base	Best	$\Delta R@1$	Range
η_{base}	10^{-3}	10^{-3}	+0.0	25.2
N_G ^a	3	3	+0.0	20.1
η_{bb}	10^{-5}	10^{-4}	+9.5	19.6
Backbone	ViT-B/16	ViT-L/16	+2.1	19.0
E	50	100	+6.1	14.7
P	16	8	+0.8	10.6
p_{drop}	0.1	0.0	+2.7	7.9
K	4	2	+0.8	7.2
λ_t	0.5	0.0	+5.1	5.8
s_a	32.0	32.0	+0.0	5.4
Image size	256	256	+0.0	5.0
m_a	0.5	1.0	+1.7	3.5
W	5	3	+2.7	2.7
d_e	256	256	+0.0	1.8
m_t	0.3	0.3	+0.0	1.8
G_{max}	1.0	1.0	+0.0	1.7
γ	0.75	0.65	+1.2	1.2
P_{stop}	10	20	+0.2	0.9 ^b
λ_w	10^{-4}	10^{-5}	+0.8	0.8

alters the query/gallery composition, so its rows are not evaluated on an identical query set. ^aEarly stopping never triggers for $P_{\text{stop}} \geq 10$ at $E = 50$.

best setting, improving Rank-1 to 83.4%; ArcFace alone is sufficiently discriminative at this dataset scale, and the triplet term interferes with optimisation. This also removes the justification for PK-sampling, explaining why $K = 2$ outperforms $K = 4$. Second, performance degrades monotonically with dropout, and weight decay prefers its smallest tested value. Together with the absence of overfitting at $E = 50$, this indicates that the model is not in an overfitting regime at all: the self-supervised DINOv3 features and the augmentation pipeline already act as strong implicit regularisation, and explicit regularisation only discards useful capacity.

D. Combined Configuration and Sub-Additivity

To test whether parameter effects are additive, a combined configuration adopting all 12 settings with positive Δ was trained (Table IV).

TABLE IV
BASELINE VS. COMBINED CONFIGURATION

Configuration	Cosine		+Re-rank	
	R@1	mAP	R@1	mAP
Baseline	77.4	83.6	78.3	84.9
$\eta_{\text{bb}} = 10^{-4}$ only	—	—	87.8	91.6
Combined (12 changes)	89.1	92.0	90.4	93.5

The combined configuration gains +12.1 pp Rank-1 and +8.6 pp mAP over the baseline, a substantial improvement. The comparison against the additive prediction, however, is the more informative result. Summing the individual Δ values gives +33.7 pp, predicting $78.3 + 33.7 = 112.0\%$ Rank-1 — above the theoretical maximum. The equivalent mAP

calculation predicts 105.6%. Only 36% of the predicted Rank-1 gain and 42% of the predicted mAP gain materialise; the consistency of these ratios across two metrics indicates a systematic property rather than an artefact.

The explanation is overlap. Several improvements address the same underlying limitation and therefore cannot be summed: η_{bb} , E , W , the larger backbone, and γ all give the backbone more capacity or more opportunity to adapt to the turtle domain, while $\lambda_t = 0$, $p_{\text{drop}} = 0$, and $\lambda_w = 10^{-5}$ all relax regularisation on a model that was never overfitting. Consistently, raising η_{bb} alone already recovers +9.5 pp, i.e. about 78% of the combined gain; the other eleven changes contribute +2.6 pp between them.

The methodological consequence is that OFAT usefully ranks parameters by importance and direction but cannot predict combined performance, and stacking all OFAT winners does not guarantee an optimum. One case is visible in our own selection: $\gamma = 0.65$ entered the combined configuration on Rank-1 (+1.2 pp) despite reducing mAP and Rank-5 in its own ablation.

E. Cross-Dataset Generalisation

The combined checkpoint (ViT-L/16 DINOv3) was applied to SeaTurtleID2022 [9] without retraining. Twenty-five identities were drawn at random as registered, with $N_G = 3$ gallery images each and 217 queries. The inference protocol is identical to the in-distribution evaluation, so any difference is attributable to domain change alone.

TABLE V
CROSS-DATASET RETRIEVAL ON SEATURTLEID2022 (NO RETRAINING)

Metric	In-distribution	SeaTurtleID2022
Rank-1 (%)	90.4	81.1
Rank-5 (%)	96.6	97.2

Rank-1 falls by 9.3 pp, so transfer is not free, but the degradation is far from catastrophic: Rank-5 slightly exceeds the in-distribution value, and Rank-1 still surpasses the baseline configuration on its own native domain (78.3%). The fine-tuned DINOv3 embedding therefore generalises across capture conditions.

F. Open-Set Identification with Rejection

Field images may show unregistered animals. Adding five identities absent from the gallery brings the query set to 266 (217 registered, 49 unregistered). We compare two structurally different rejection rules. The *global threshold* is absolute: the top-1 cosine score is compared against a single tuned value (maximising Youden’s J). The *background class* is relative: prototypes of turtles outside the gallery are retained as multiple vectors, and a query is accepted when $s = s_{\text{reg}} - s_{\text{bg}} \geq \text{margin}$, where s_{bg} is the maximum similarity to any background prototype. Because s_{bg} is recomputed per query, the rule normalises hub identities. Background prototypes come from a disjoint image source, so both schemes are evaluated on the identical 266-query set.

TABLE VI
IDENTIFICATION WITH REJECTION ON SEATURTLEID2022

Metric	Global threshold	Background class
Decision rule	absolute	relative
Identification accuracy (%)	63.5	64.3
Registered Rank-1 (%)	55.3	56.7
False-reject rate (%)	43.4	41.9
Unregistered recall (%)	100.0	98.0
False-accept rate (%)	0.0	2.0

Registered Rank-1 collapses from 81.1% to 55.3% and 56.7%. The ranking model is unchanged between Table V and Table VI; only a rejection stage placed before ranking has been added, and it discards over 40% of valid registered queries.

The decisive observation is that both rules land on nearly the same operating point — every difference is on the order of one or two queries — despite being mechanically unrelated. The cause is visible in the score distributions. All top-1 scores, registered and unregistered alike, are confined to a narrow band of roughly 0.85–0.98: because every image shares near-identical structure (a turtle head against an underwater background), embeddings occupy a narrow cone of the hypersphere and any two images are already similar before identity is considered. The unregistered distribution is not merely shifted but embedded inside the lower tail of the registered distribution, so any threshold high enough to exclude all unregistered queries necessarily removes a large share of registered ones. Separability is only moderate, with ROC-AUC of 0.77–0.79, which bounds every possible decision rule. For the relative scheme, the two classes overlap almost completely along the s_{bg} axis, so s_{bg} carries no discriminative information and $s = s_{reg} - s_{bg}$ is effectively s_{reg} plus noise.

Qualitative inspection confirms that the representation is not at fault. Among the five most confident errors, all are registered queries wrongly rejected — none is a false accept — and in every case the correct identity remains within the top five, at rank 1 for three of them (scores 0.853, 0.867, 0.868). Retrieval had already placed the right answer first; the rejection stage discarded it. Among accepted registered queries, retrieval accuracy is $0.567/(1 - 0.419) \approx 97.6\%$ for the background class and $\approx 97.7\%$ for the threshold, i.e. both rules are simply strict enough to filter the hard cases out in advance.

G. Comparison with Prior Work

Table VII is contextual only: the datasets, identity counts, and protocols differ, and the closed-set 95.3% of [10] on 20 identities is not commensurable with open-set evaluation on held-out identities. The comparison’s value lies in its last three rows, which show the same model losing 9.3 pp to domain transfer and a further 25.8 pp to the requirement of rejecting unknowns — a cost that closed-set reporting renders invisible.

TABLE VII
CONTEXT AGAINST PRIOR WORK (NOT DIRECTLY COMPARABLE)

Method	Dataset	Protocol	R@1 (%)
ResNet-50 [10]	20 identities	closed-set	95.3
Adam et al. [9]	SeaTurtleID2022	open-set	86.8
Ours (combined)	SeaTurtleIDHeads	open-set	90.4
Ours (transfer)	SeaTurtleID2022	open-set	81.1
Ours (w/ rejection)	SeaTurtleID2022	open-set	55.3

V. DISCUSSION AND LIMITATIONS

Our results separate two failure sources that are usually conflated. The embedding generalises: it survives a change of dataset with a measured and moderate loss, and it keeps the correct identity at rank 1 even in the cases the system reports most confidently and wrongly. What fails is the mapping from similarity scores to a registered/unregistered decision. This suggests that future work on wildlife Re-ID deployment should target score calibration and domain-specific score normalisation rather than further retrieval accuracy: an extra point of Rank-1 is worth little when 42% of valid queries are rejected before ranking.

Several limitations bound these conclusions. All results come from a single seed, so parameters whose Rank-1 range is below 2 pp cannot be distinguished from random fluctuation; multi-seed means and standard deviations are required. The OFAT baseline sits in an under-trained regime ($E = 100$ still improves over $E = 50$), so other parameters were assessed at a non-converged operating point. The rejection threshold was tuned on the same set it was evaluated on and only 49 unregistered queries were available, so the reported 100% unregistered recall should not be read as a guarantee on new data. Finally, head detection and segmentation are assumed rather than integrated.

VI. CONCLUSION

We presented a sea turtle Re-ID system built on a DINOv3-pretrained Vision Transformer, trained under an open-set identity split on SeaTurtleIDHeads. A convention-derived baseline reaches 78.3% Rank-1 and 84.9% mAP, and a tuned configuration reaches 90.4% and 93.5%. A sensitivity study over 19 parameters shows that the backbone learning rate dominates, contributing roughly 78% of the total gain, that ArcFace alone outperforms the conventional ArcFace-plus-triplet combination at this data scale, and that OFAT gains are strongly sub-additive and must not be summed. The model transfers to SeaTurtleID2022 without retraining at 81.1% Rank-1. Under rejection of unregistered individuals, however, Registered Rank-1 falls to 55.3–56.7% at a false-reject rate near 42%, and two structurally different decision rules converge on the same operating point because top-1 scores of registered and unregistered queries are only moderately separable (ROC-AUC 0.77–0.79). The bottleneck for field deployment is score calibration, not representation quality. Future work should pursue multi-seed evaluation, threshold

tuning on a held-out validation set with a larger unregistered population, exploitation of left-right profile similarity [12], and integration of head detection into an end-to-end pipeline.

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